Noise + Light Dual Logger

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Chapter 1

Purpose, Features, and Ratings

1.1 Purpose

The purpose of this project is to build a dual-channel monitoring system that captures acoustic and lighting conditions simultaneously, time-aligns both data streams, and sends synchronized measurements to a mobile app through Bluetooth communication interface for correlated visualization and analysis. The design prioritizes interference reduction, stable signal processing, and reliable synchronization to ensure accurate environmental monitoring. The interference reduction is achieved by means of proper distance among different components. Decoupling capacitors are introduced to decrease the supply noise affecting the components.

1.2 Key Features

- Feature 1: Indoor Applications even in low light conditions
- Feature 2: Extreme Reliability
- Feature 3: Rechargeable battery
- Feature 4: Standard Bluetooth communication interface
- Feature 5: 3 different color LED signaling for correct or wrong behavior
- Feature 6: Measurement of the sound level of the environment
- Feature 7: Measurement of the environment light level
- Feature 8: Synchronous operation on both sound and light information
- Feature 9: Data filtering and processing

1.3 Ratings and operating conditions

In the following, key parameters that are useful for the final user are reported, to understand the system operating conditions.

- normal operating temperature:from - 40 °C to 85 °C, for all components

- operating temperature while charging: from 0 °C to 60 °C
- acoustic sensitivity frequency range: from 60 Hz to 20 kHz, matching the human ear sensitivity.
- light measurement range: from 0.01 lux to 83 k lux
- central wave length of light sensor: 550 nm, matching the human eye sensitivity
- Battery life: 5 Days
- full battery recharge time:approximately 20 hours

Chapter 2

Block Diagrams

2.1 System Overview

The system is a battery-powered, dual-sensor acquisition platform designed to measure acoustic and illumination conditions simultaneously and forward time-aligned data to a mobile application via Bluetooth Low Energy (BLE). The architecture is organized into a power-management chain, a central processing unit (MCU) with timing references, and two sensing channels (acoustic and light) that are sampled and synchronized in firmware. The firmware itself is run by the MCU, which is Texas Instruments CC2640R2FRHBR. This small embedded processor is chosen because of its low-power capabilities and its native Bluetooth interface. The power switch turns on and off the entire system, which is powered by a rechargeable lithium-ion battery. The battery is charged by means of a micro USB connection that charges at 0.5 A, 4.2 V. The MCU drives 3 LED, which are red, green and blue, to signal the correct or faulty behavior. In figure 2.1, the block diagram of the system is represented.

2.1.1 Functional Units

- **Battery charger subsystem:** This sub-block handles power management and charging for the system. It features an Analog Devices MAX1555 standalone charger, which manages the charging cycle for the battery connected via the BAT1 terminal. Power is delivered to the charger through a J1 micro-USB connector, serving as the primary input source.

- **LDO subsystem:** This sub-block provides a regulated 3.0 V supply to the system using the Texas Instruments TLV75801PDBVR LDO. Power distribution is controlled by switch SW1, which is positioned between the battery and the regulator to serve as the system's primary power toggle.

- **MCU and RF subsystem:** the MCU manages sensor configuration, data sampling schedules, and timestamping. It also handles digital filtering and the packaging of telemetry for wireless transmission. Precise timing for real-time operations and radio synchronization is maintained by dedicated 32.768 kHz (LF) and 24 MHz (HF) crystal oscillators, while Bluetooth connectivity is facilitated via the ANT1 antenna.

- **Sensing channels:** an **acoustic sensor** captures sound-related signals, while a **light sensor** measures ambient illumination. Both channels are treated as separate signal paths to reduce mutual interference.

- **User/diagnostic interface:** a reset (RST) button provides manual recovery/control, and 3 RGB LEDs indicate operating status such as power, acquisition, or BLE connection status.

2.1.2 Power, Signal, and Control Flow

Power flows through the *battery* from the *charger* (for safe recharging), then through the *power switch* and *voltage regulator* to supply the MCU, sensors, and indicators. The MCU provides the primary *control flow* by configuring the sensors, triggering/clocking acquisition, and enforcing synchronization between the acoustic and light measurement streams. The *signal flow* originates at the acoustic and light sensors, is acquired by the MCU (with time alignment and digital filtering), and is then transmitted over BLE to the mobile application for correlated trend visualization.

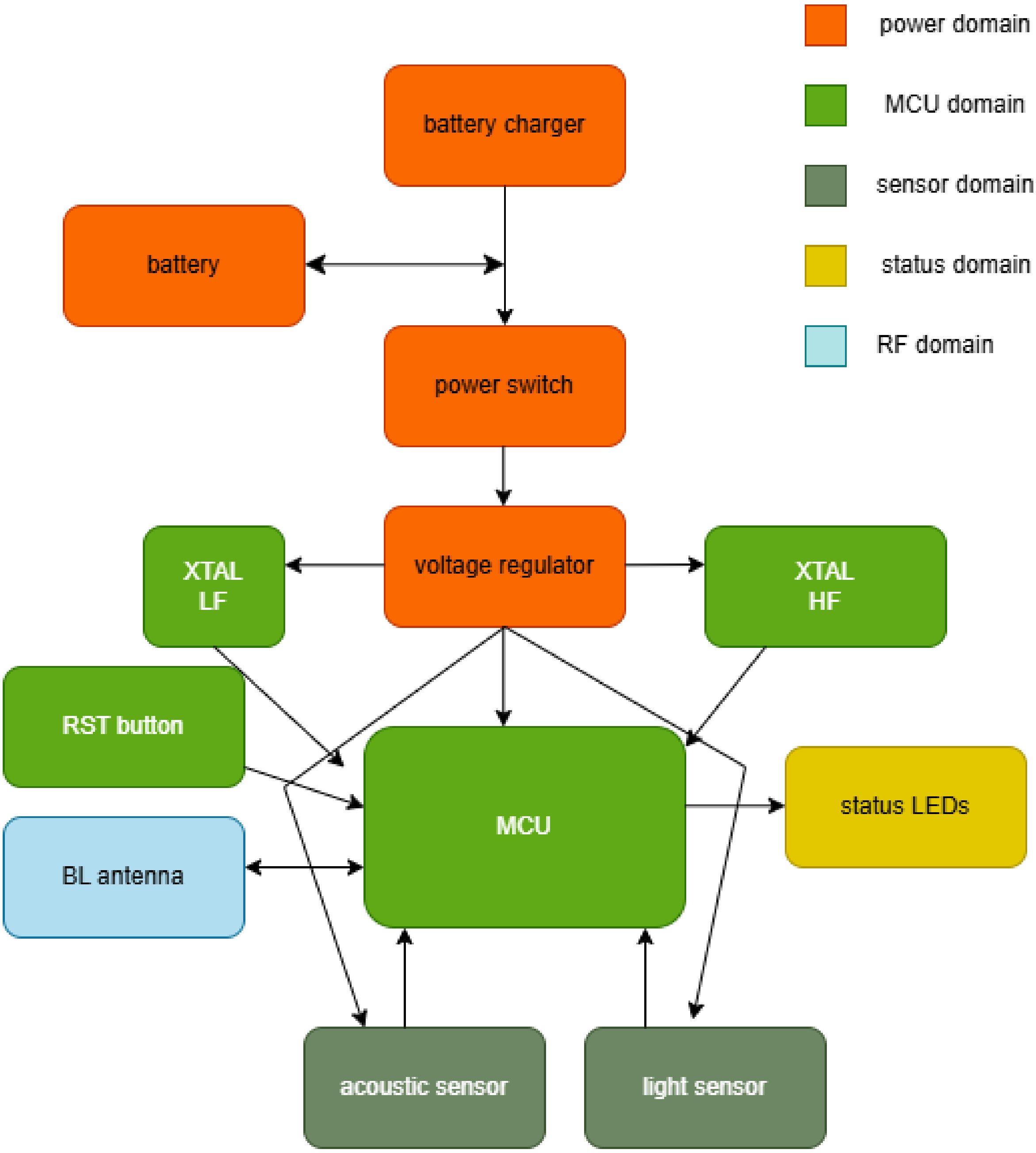

Figure 2.1: System block diagram

Chapter 3

Design Choices

This chapter summarizes the main component and architecture choices for the EchoLight system, and explains why they were selected compared to alternative options evaluated during the design phase.

3.1 Antenna Choice: Chip Antenna vs PCB Antenna

Decision: Use a ready-made chip antenna (e.g., RFANT3216120A5T) and an RF matching network, instead of designing a PCB-printed antenna.

Why:

- **Lower design risk:** PCB antennas require strict layout rules, controlled stack-up, and accurate keepout/ground clearances. Small deviations can detune the antenna and reduce BLE range.
- **Predictable RF performance:** A commercial chip antenna has characterized behaviour and reference layouts, which improves the probability of meeting communication requirements on the first PCB revision.
- **Simpler validation:** With a chip antenna, the main RF tuning work is concentrated in the matching network and placement rules, rather than optimizing the antenna geometry itself.

Trade-off: Chip antennas can be slightly more expensive and may be less efficient than a well-designed PCB antenna, but they significantly reduce integration complexity and uncertainty for a compact prototype.

3.2 Sensor Selection

3.2.1 Light Sensor: OPT3001 vs Alternatives

Decision: Use the OPT3001 ambient light sensor.

Why:

- **Direct lux measurement and digital interface:** the sensor provides a calibrated digital output over I^2C, which simplifies integration and reduces analog sensitivity in a noisy environment.

- **Wide dynamic range:** suitable for indoor monitoring, including low-light conditions.
- **Low-power operation:** supports periodic sampling with low average current, matching the battery-powered goal.

Why not the other options: alternatives were not selected when they offered unnecessary features for the target use-case (e.g., higher complexity or higher power), or when their integration requirements were less compatible with a compact, low-risk PCB prototype.

3.2.2 Acoustic Sensor: Digital MEMS (ICS-43434) vs Alternatives

Decision: Use a digital MEMS microphone with an I^2S output (e.g., ICS-43434).
Why:

- **Noise robustness:** digital audio reduces susceptibility to PCB coupling compared to purely analog microphone signals, supporting stable multi-channel acquisition.
- **Deterministic sampling:** I^2S provides a well-defined clocked stream, which helps time-align acoustic features with the light sensor measurements.
- **Feature-based processing:** the system only requires compact acoustic descriptors (e.g., RMS level) rather than full audio streaming, so a digital MEMS microphone is an efficient and clean integration choice.

Why not the other options (from slides): other candidates were not chosen when they increased analog routing complexity, required additional analog front-end circuitry, or increased power and processing cost beyond what is necessary for correlated environmental monitoring. Furthermore, analog communication is susceptible to environmental noise.

3.3 Wireless and Processing Strategy

Decision: Use BLE to send synchronized light + acoustic features to a mobile application, rather than transmitting raw high-rate audio continuously.
Why:

- **Battery life:** compact packets reduce radio-on time and average current consumption.
- **Synchronization:** the MCU can enforce a fixed acquisition period and time-align both sensing channels before transmission.
- **Usability:** a phone application can visualize correlated trends without requiring large data storage or bandwidth.

3.4 Summary

The selected antenna and sensors prioritize **low integration risk**, **stable measurements**, and **predictable performance** on the first PCB revision. Component choices were guided by the project goals: synchronized environmental monitoring (sound + light), low power operation, and reliable BLE communication to a mobile application.

Chapter 4

Key Design Equations or Relations

4.1 Time Alignment and Sampling Budget

The system generates one synchronized measurement point per second by using a **center-aligned snapshot**: audio is sampled over a short window centered within the light sensor integration period. This provides time alignment while minimizing power and memory usage. Worst-case light conversion time is approximately $T_{\text{light}} \approx 883.00\,\text{ms}$ (dark conditions), and the audio snapshot window is $T_w = 100.00\,\text{ms}$ within the light integration interval.

4.1.1 Audio buffer sizing

Given sampling rate f_s and snapshot duration T_w:

$$N = f_s \cdot T_w \tag{4.1}$$

For the selected high-power microphone mode:

$$N = 51.60\,\text{kHz} \cdot 0.10\,\text{s} = 5160 \text{ samples} \tag{4.2}$$

If samples are stored as 16-bit words, the required buffer size is:

$$M = N \cdot \frac{16}{8} = 5160 \cdot 2 = 10320 \text{ bytes} \approx 10.32\,\text{kB} \tag{4.3}$$

4.1.2 Processing time estimate (IIR + RMS)

Using a cycle budget of C_s CPU cycles per sample and CPU clock f_{CPU}:

$$T_{\text{proc}} = \frac{N \cdot C_s}{f_{\text{CPU}}} \tag{4.4}$$

In the project estimation, $C_s \approx 100$ cycles/sample yields $T_{\text{proc}} \approx 10.75\,\text{ms}$ for the 100 ms audio snapshot.

4.1.3 BLE payload sizing

Each synchronized report contains:

$$\text{Payload} = 2 \text{ bytes (Audio RMS)} + 2 \text{ bytes (Light)} = 4 \text{ bytes} \tag{4.5}$$

Including protocol overhead, the transmitted packet size is estimated at ≈ 20 bytes, with a conservative radio active time of ≈ 2–3 ms at $1.00\,\text{Mbps}$.

4.2 Average Current and Battery Life Estimation

To compare design options and validate the expected autonomy, each subsystem average current is computed using a duty-cycle weighted relation:

$$I_{\text{avg}} = \frac{T_{\text{active}} I_{\text{active}} + T_{\text{sleep}} I_{\text{sleep}}}{T_{\text{total}}} \tag{4.6}$$

4.2.1 Subsystem average currents

Example values used in the project (per 1 second cycle) include:

- Light sensor average current computed with Equation (4.6) (worst-case integration).
- Microphone average current computed with Equation (4.6) for $T_{\text{active}} = 100.00\,\text{ms}$.
- MCU + BLE average current based on processing + TX + standby time weighting.
- Sensor controller + DMA + I2S average current based on wakeups and active windows.
- LED average current using pulse duty cycle:

$$I_{\text{LED}} = \frac{T_{\text{pulse}} \cdot I_{\text{ON}}}{T_{\text{period}}} \tag{4.7}$$

4.2.2 Battery life

Total average current is the sum of subsystem contributions:

$$I_{\text{total}} = I_{\text{Mic}} + I_{\text{Light}} + I_{\text{MCU}} + I_{\text{LED}} + I_{\text{SC+DMA+I2S}} \tag{4.8}$$

Battery life estimate:

$$T_{\text{life}} = \frac{C_{\text{bat}}}{I_{\text{total}}} \tag{4.9}$$

Using $C_{\text{bat}} = 500.00\,\text{mAh}$ and the project's estimated $I_{\text{total}} \approx 4.16\,\text{mA}$ gives:

$$T_{\text{life}} \approx \frac{500.00\,\text{mAh}}{4.16\,\text{mA}} \approx 120.35\,\text{h} \approx 5 \text{ days} \tag{4.10}$$

Chapter 5

Complete Schematics

5.1 MCU Schematic Overview

The chosen MCU is the Texas Instruments CC2640R2FRHBR. Its package is the 5 mm × 5 mm RHB VQFN32. It features 32 pins, 15 of them are GPIOs and can be connected to sensors and LEDs, to signal the correct or wrong behavior of the system. In figure 5.1, the schematic of the MCU connections is reported, including decoupling circuits and matching circuit for the antenna.

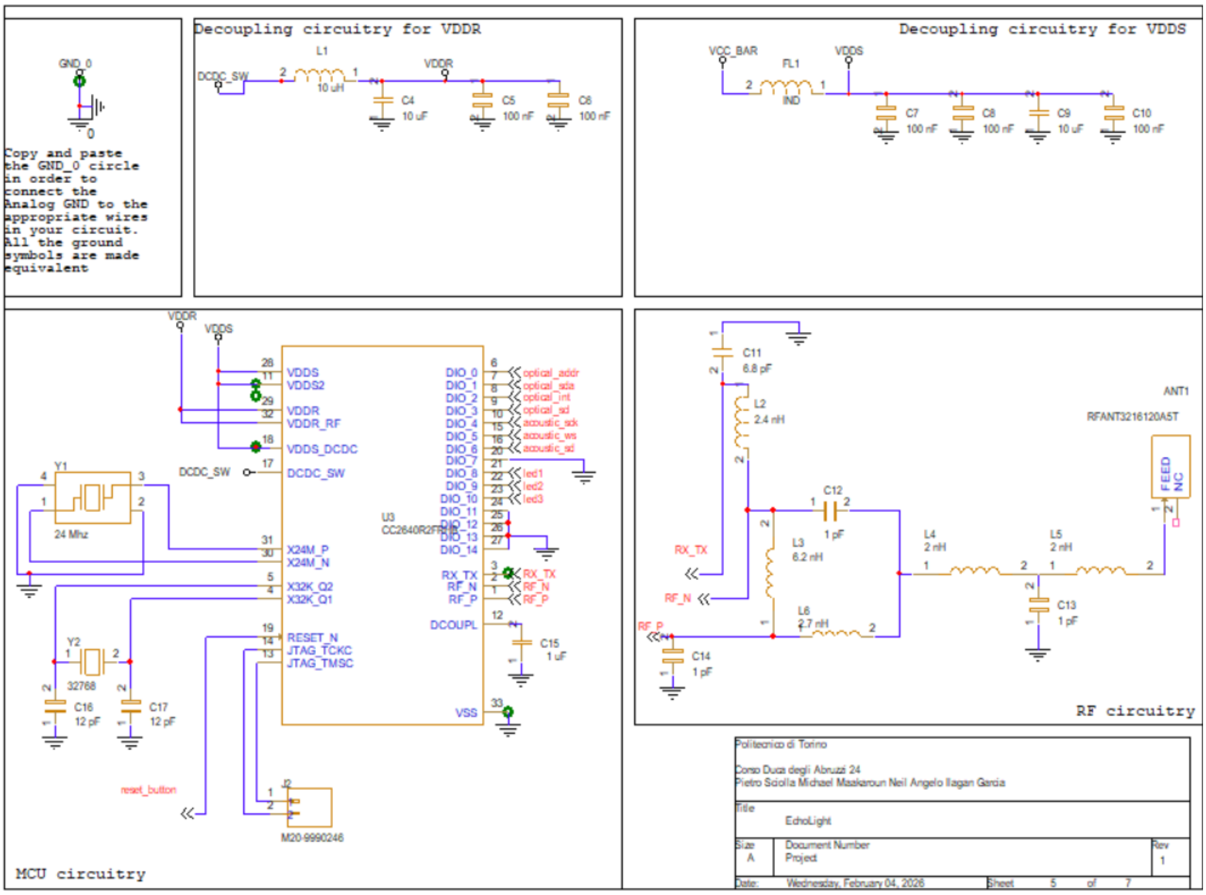

Figure 5.1: MCU Schematic

In the following table, the main characteristics of the MCU are reported.

Table 5.1: MCU main characteristic

parameter	**Value**
RefDes	U1
Part number	CC2640R2FRHBR
operating voltage	3.0 V
Active-Mode RX	5.9 mA
Active-Mode TX at 0 dBm	6.1 mA
Actice-Mode TX at 5 dBm	9.1 mA
Active-Mode MCU	48.5 CoreMark/mA
Active-Mode MCU	61 µA/MHz
Active-Mode sensor controller	0.4mA + 8.2 µA/MHz
Standby	1.1 µA
Shutdown	100 nA
Package	RHB VQFN32

In the following, the list of MCU pins, their function and their connection is presented.

- Pin 1,RF_P: the positive RF input, connected to the matching circuit of the antenna
- Pin 2, RF_N: negative input signal, connected to the matching circuit of the antenna
- Pin 3, RX_TX: RF input and output signal, connected to the antenna matching circuit.
- Pin 4, X32K_Q1: connected to the Y2 32.768 kHz oscillator and decoupling capacitor C17
- Pin 5, X32K_Q2: connected to the Y2 32.768 kHz oscillator and decoupling capacitor C16
- Pin 6,DIO_0:GPIO, sensor controller, connected to the address pin of the optical sensor
- Pin 7,DIO_1: GPIO, sensor controller, connected to optical SDA, (Serial Data)
- Pin 8, DIO_2 : GPIO, sensor controller, connected to optical int, which is the interrupt output signal of the optical sensor
- Pin 9, DIO_3: GPIO, sensor controller, connected optical SCL, which is the serial clock of the optical sensor
- Pin 10, DIO_4: GPIO, sensor controller, connected to acoustic sck, which is the clock signal of the acoustic sensor
- Pin 11,VDDS2:1.8-V to 3.8-V GPIO supply, connected to the VDDS decoupling circuit
- Pin 12, DCOUPL:1.27-V regulated digital-supply decoupling, connected to C15
- Pin 13, JTAG_TMSC: connected to J2 header, for external programming and debugging
- Pin 14, JTAG_TCKC: connected to J2 header, for external programming and debugging
- Pin 15,DIO_5: GPIO and JTAG TDO(Test Data Output), connected to acoustic WS, which is the word select signal for the I2C interface of the acoustic sensor

- Pin 16, DIO_6: GPIO and JTAG TDI(Test Data Input), connected to acoustic SD, which is the synchronous data signal for the acoustic sensor
- Pin 17, DCDC_SW: output from internal DC/DC converter, connected to VDDR decoupling circuit
- Pin 18, VDDS_DCDC: 1.8-V to 3.8-V DC/DC supply, connected to the dedicated VDDR circuit
- Pin 19, RESET_N: reset signal for the MCU, active low, connected to the reset button
- Pin 20, DIO_7: GPIO, connected to ground
- Pin 21, DIO_8: GPIO, connected to the negative pin of the red LED
- Pin 22, DIO_9: GPIO, connected to the negative pin of the blue LED
- Pin 23, DIO_10: GPIO, connected to the negative pin of the green LED
- Pin 24, DIO_11: GPIO, connected to ground
- Pin 25, DIO_12: GPIO, connected to ground
- Pin 26, DIO_13: GPIO, connected to ground
- Pin 27, DIO_14: GPIO, connected to ground
- Pin 28, VDDS: 1.8-V to 3.8-V main chip supply, connected to VDDS decoupling circuit
- Pin 29, VDDR: 1.7-V to 1.95-V supply, connected to VDDR decoupling circuit
- Pin 30, X24M_N: 24 MHz crystal oscillator connection to pin 1
- Pin 31, X24M_P: 24 MHz crystal oscillator connection to pin 2
- Pin 32, VDDR_RF: 1.7-V to 1.95-V supply, connected to VDDR decoupling circuit
- Pin 33, EGP: thermal pad, connected to ground

5.1.1 Power and Decoupling

The MCU is supplied through the digital supply rails (VDDS/VDDR and VDDR_RF), each locally decoupled using capacitors placed close to the corresponding pins.
The decoupling circuit for VDDS, taken from the MCU data sheet, comprises ferrite-bead FL1, C7, C8, C9 and C10. Those capacitors are connected in parallel.
The decoupling circuit for VDDR, taken from the MCU data sheet, consists of L1, C4, C5 and C6. Those capacitors are connected in parallel.
DCDC_SW is connected to the inductor L1.

5.1.2 Clock Sources

Two crystals are used to satisfy timing accuracy requirements:

- **HF crystal (24 MHz):** provides the main high-accuracy reference for RF communication and system operation.
- **LF crystal (32.768 kHz):** provides a low-power time base for sleep timing and accurate periodic wake-up scheduling.

Each crystal is implemented with its recommended load capacitors to match the target load capacitance and ensure stable oscillation.

5.1.3 RF Matching and Antenna Interface

The RF output pins are connected to the antenna through an LC matching network.
This circuit comprises L2, L3, L4, L5, L6, C11, C12, C13, C14.
The values of those components and their connections are taken from the MCU datasheet.
This network is responsible for impedance transformation and harmonic/noise filtering to maximize RF power transfer, improve link stability, and satisfy emissions constraints.
The unconnected pin of the antenna is just for mechanical stability.

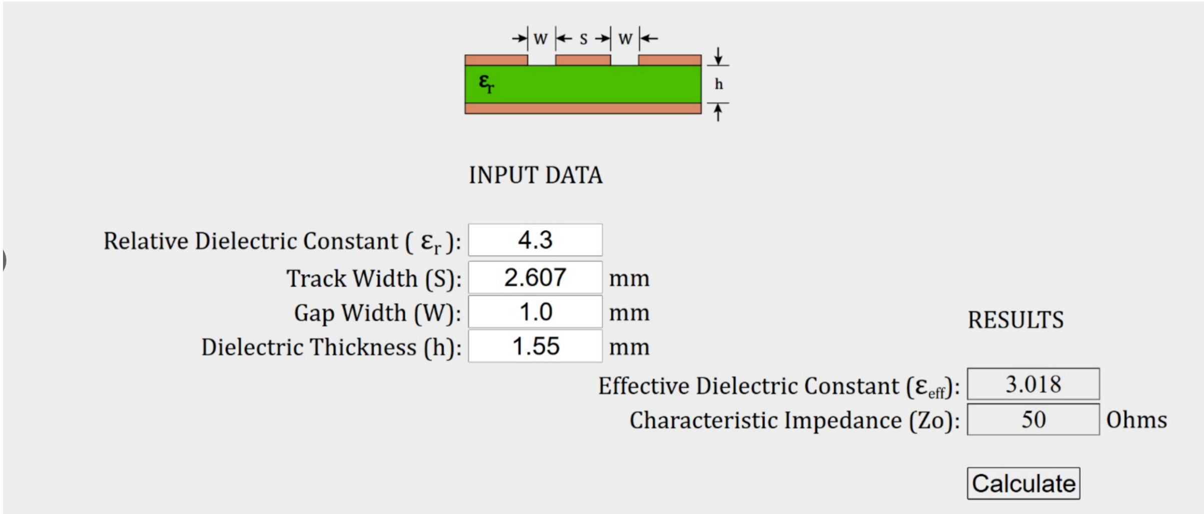

Figure 5.2: Microstrip width calculation for the RF trace.

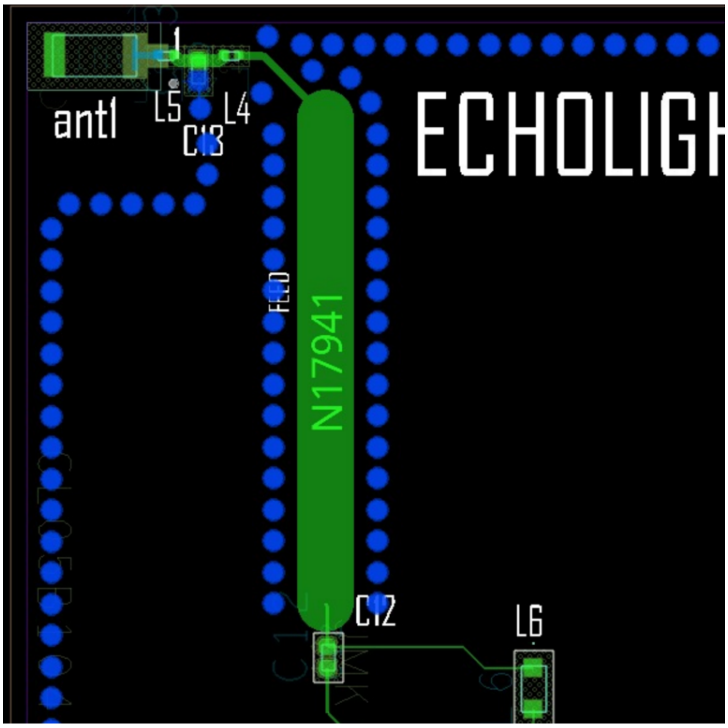

Figure 5.3: RF matching network and controlled-impedance trace placement.

The RF trace width was calculated from the PCB stack-up to achieve a controlled impedance (typically $50.00\,\Omega$). The matching components were then placed close to the MCU and the trace was routed as short and smooth as possible to minimize parasitics and maintain impedance continuity to the antenna.

5.1.4 Reset, Programming, and I/O Interfaces

A dedicated reset input is connected to a push-button for manual recovery. A debug/programming header exposes the JTAG signals for firmware flashing and debug. The JTAG pins are connected to the dedicated header J2
GPIO lines are allocated for the acoustic interface signals, optical sensor control/data lines, and status LEDs, enabling acquisition control, synchronization, and user feedback.

5.2 Battery Charger

The device is powered by the UWN18650C1-2000mAh single-cell Lithium-ion rechargeable battery (net `V_BAT`). Its capacity is equal to 2000 mAh. The battery is held by the connector BAT1 in the schematic A linear charger (MAX1555) enables its charging from a USB input connector.
The schematic for the connections of the battery charger and the battery itself is reported in 5.4. The table 5.2 lists the main characteristics of the MAX1555 battery charger. The chosen battery charger operates without the need of external FETs or diodes.
The MAX1555 can charge from either the USB input or the DC input. The battery does not charge from both sources at the same time. The MAX1555 automatically detects the active input and charges from that. If both power sources are active, the DC input takes precedence.

The MAX1555 feature a precharge current to protect deeply discharged cells. If VBAT is less than 3V, the device enters precharge mode where charging current is limited to 40mA.

The following list reports the pins of the battery charger, their function and their connection:

- Pin 1, USB: USB Port Charger Supply Input, connected to the VDD pin of J1 USB connector
- Pin 2, GND: connected to ground
- Pin 3, CHG': Active-Low Open-Drain Charge Status Indicator.CHG' pulls low when the battery is charging. CHG' goes to a high-impedance state, indicating the battery is fully charged. Unconnected
- Pin 4, DC: DC Charger Supply Input for an AC Adapter. connected to ground
- Pin 5, BAT: Battery connection

Table 5.2: battery charger characteristics

Parameter	**Value**
RedDes	U1
Type	Battery charger
Part number	MAX1555EZKT
USB input voltage	4.2 V
Battery voltage	3.7 V
Dropout Voltage	60 mV
USB Supply Current	1.65 mA
USB Charging Current	80 mA
Maximum Battery Leakage Current	5 µA
Package	SOT-23

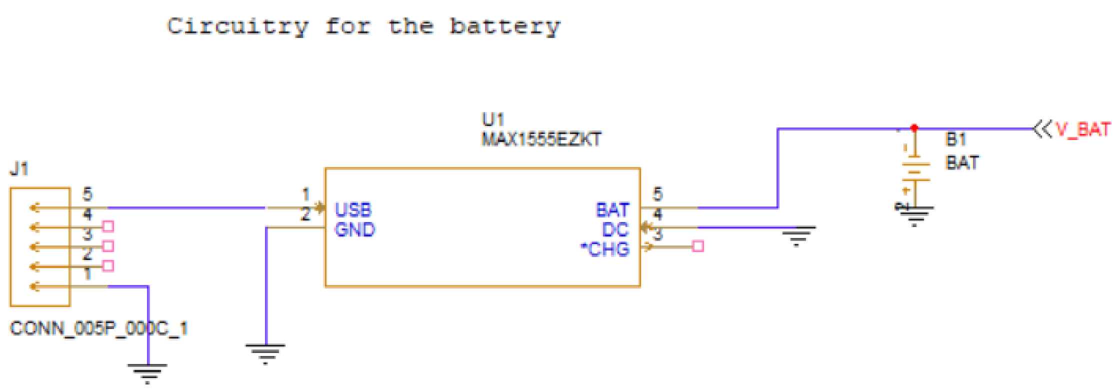

Figure 5.4: Battery Charger Schematic

5.2.1 Battery charging time estimate (MAX1555)

A first-order charge time estimate can be obtained from:

$$t_{\text{charge}} \approx \alpha \frac{C_{\text{BAT}}}{I_{\text{CHG}}} \tag{5.1}$$

where C_{BAT} is battery capacity (Ah), I_{CHG} is charge current, and $\alpha \approx 1.2$ accounts for constant-current/constant-voltage charging overhead. This provides an approximate charging duration from USB.

5.3 LDO

For this project, the Texas Instruments TLV75801PDBVR LDO voltage regulator is chosen. This device is configured to deliver an output voltage equal to 3.0 V.
In figure 5.5 the schematic of the LDO subcircuit is reported. R1 and R2 are used to adjust the output voltage of the regulator to 3.0 V. C3 and C2 are introduced for decoupling purposes. The switch SW1 connects the input pin of the LDO to either V_BAT (output pin of the battery) or to ground. In table 5.3, the main characteristics are presented.

Table 5.3: LDO characteristics

Parameter	**Value**
RefDes	U2
Type	LDO voltage regulator
Part number	TLV75801PDBVR
Feedback voltage	0.55 V
Output accuracy	0.7 %
Line regulation	0.2 mV
Load regulation	0.030 V/A
Output current limit	720 mA
Short-circuit current limit	350 mA
Input voltage	3.7 V
Output Voltage	3.0 V
Dropout Voltage	95 mV
USB Supply Current	1.65 mA
USB Charging Current	80 mA
Maximum Battery Leakage Current	5 µA
Package	SOT-23-5

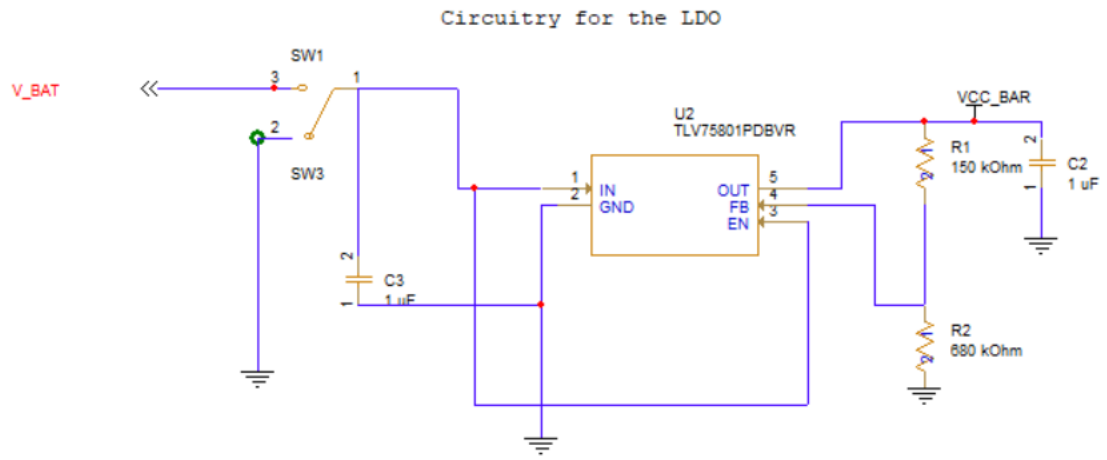

Figure 5.5: LDO Schematic

The following list reports the pin function and their connection.

- Pin 1, OUT: Regulated output voltage pin, connected to C2, VCC_BAR (the global voltage supply of the circuit) and R1
- Pin 2, FB: input to the control loop error amplifier and is used to set the output voltage of the LDO. Connected to R1 and R2
- Pin 3, GND: ground pin, connected to ground
- Pin 4, EN: Enable pin. Drive EN greater than VEN(HI) to turn on the regulator. Connected to the switch SW1.
- Pin 5, DNC: do not connect

5.3.1 Adjustable LDO output voltage (TLV75801)

The TLV75801 uses a feedback divider (R2, R1) to set the output voltage:

$$V_{\mathrm{OUT}} \approx V_{\mathrm{REF}}\left(1 + \frac{R2}{R1}\right) \tag{5.2}$$

where V_{REF} is the internal reference voltage of the adjustable LDO. From Equation (5.2), the resistor ratio is:

$$\frac{R2}{R1} \approx \frac{V_{\mathrm{OUT}}}{V_{\mathrm{REF}}} - 1 \tag{5.3}$$

Example (if $\mathrm{V_{OUT}} = 3.00\,\mathrm{V}$ *and* $V_{\mathrm{REF}} = 0.55\,\mathrm{V}$):$\frac{R2}{R1} = \frac{3.0}{0.55} - 1 = 4.45$(5.4) so one valid choice is $R1 = 150.00\,\mathrm{k\Omega}$ and $R2 = 680.00\,\mathrm{k\Omega}$.

5.3.2 Dropout constraint

To guarantee regulation, the input must satisfy:

$$V_{\mathrm{IN}} \geq V_{\mathrm{OUT}} + V_{\mathrm{DO}} \tag{5.5}$$

where V_{DO} is the LDO dropout voltage at the expected load current. This equation is used to verify that `V_BAT` remains high enough (over the battery discharge range) for the chosen `VCC_BAR`.

5.3.3 LDO power dissipation (thermal check)

Since an LDO is linear, its approximate power loss is:

$$P_{\mathrm{LDO}} \approx (V_{\mathrm{IN}} - V_{\mathrm{OUT}})\, I_{\mathrm{LOAD}} \tag{5.6}$$

This relation is used to estimate regulator heating in the worst case (high `V_BAT` and maximum load current).

5.4 Acoustic sensor schematic

The chosen acoustic sensor is the Invensense ICS-43434. It is a digital sensor with I2S interface. The table 5.4 reports the main characteristics of the device. The conenctions of the device are in figure 5.6.

In the following, the pins of the acoustic sensor, their function, and their connections are presented.

- Pin 1, WS: Serial Data-Word Select for I2S Interface. Connected to the dedicated pin of the MCU
- Pin 2, LR: Left/Right channel select. When set low, the microphone outputs its signal in the left channel of the I^2S frame. When set high, the microphone outputs its signal in the right. Connected to ground
- Pin 3, GND: ground. connected to ground
- Pin 4, SCK: Serial Data Clock for I2S, connected to the dedicated pin of the MCU
- Pin 5, VDD: Powe pin. Connected to VCC_BAR through C1 capacitor
- Pin 6, SD: Serial Data Output for I^2S Interface. Connected to the dedicated pin of the MCU channel.

Table 5.4: Acoustic sensor characteristics

Parameter	**Value**
RefDes	MIC1
type	light sensor
High performance sensitivity	-26 dB
Low-Power	-26
High performance SNR	65 dbA
Low-Power SNR	64 dBA
High performance current	490 µA
Low-Power current	230 µA
High performance Sample rate	23 – 51.6 kHz
Low-power Sample rate	6.25 – 18.75 kHz
PSRR	-100 dB
THD	0.2 %

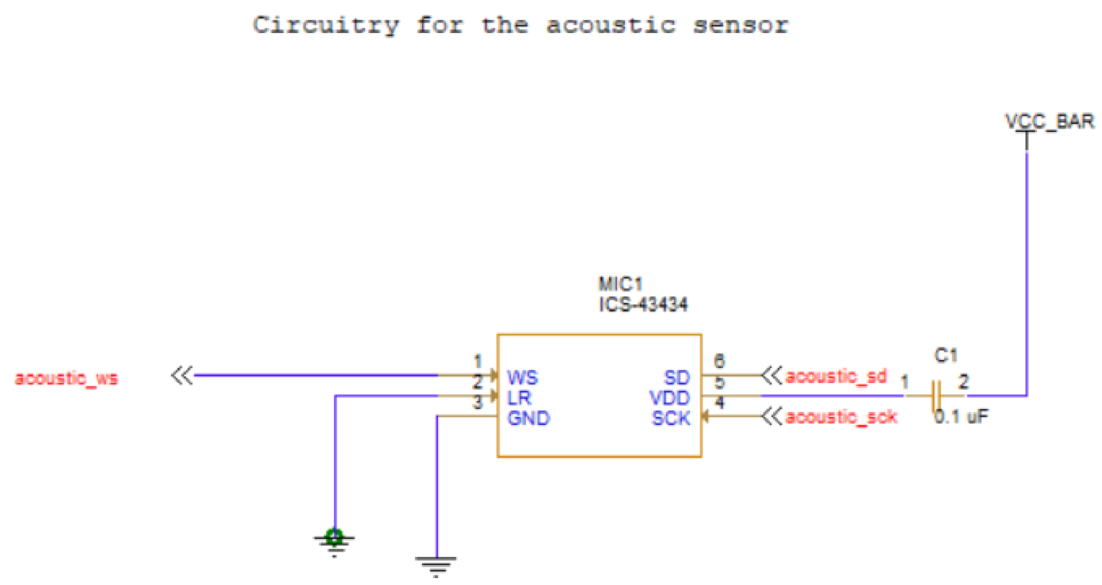

Figure 5.6: Acoustic sensor schematic

5.5 Optical Sensor Schematic

The optical sensor chosen is the Texas Instruments OPT3001DNPR, which features an I2C interface.
Its main characteristics are reported in table 5.5. The description of the connections of the optical sensor is given in figure 5.7

Table 5.5: Status LED main characteristics

Parameter	**Value**
RefDes	U4
Type	Optical Sensor
Part number	OPT3001DNPR
Peak irradiance spectral responsivity	550 nm
Resolution	0.1 lux
Full-scale illuminance	83865.6 lux
PSRR	0.1 %/V
Input current	0.01 µA

The list of the pins, their function and their connection is reported in the following list.

- Pin 1, VDD: Device power, connected to VCC_BAR, output of the LDO voltage regulator
- Pin 2, ADDR: Address pin of the optical sensor. It is dedicated to the dedicated pin of the MCU
- Pin 3, GND: ground, connected to ground

- Pin 4, SCL: clock for the I2C interface, connected to the dedicated MCU pin and pulled up by R6
- Pin 5, INT: Interrupt output, connected to the dedicated pin of the MCU, pulled up by R8
- Pin 6, SDA: serial data for the I2C interface. Connected to the dedicated pin of the MCU and pulled up by R7
- Thermal pad, connected to ground

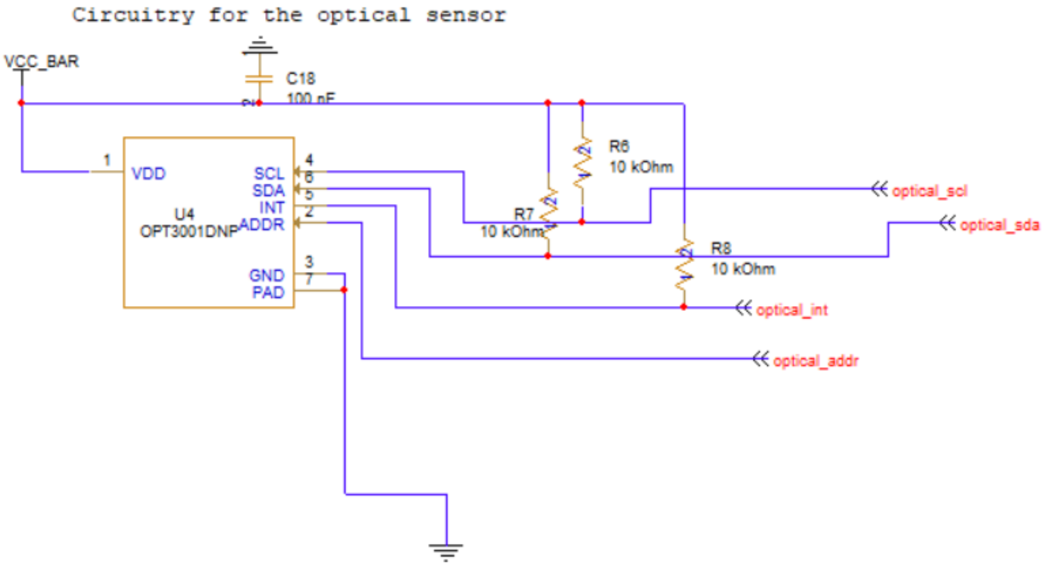

Figure 5.7: Optical sensor schematic

5.6 Status LED Schematic Overview

The system uses a RGB status LED (LED2: XZCMECBDDGK53W) to provide visual feedback during operation (power on, BLE connection, acquisition status and error conditions). The LED is driven by MCU GPIO pins through current-limiting resistors, allowing software-controlled indication patterns while keeping the average current low.

In Figure 5.8, the schematic of the LED connection is reported. In the following table, the main characteristics of the LED are summarized.

Table 5.6: Status LED main characteristics

Parameter	**Value**
RefDes	LED1
Type	RGB LED
Part number	XZCMECBDDGK53W
Drive method	MCU GPIO (through series resistors)
Current limit method	External resistors (R_R, R_G, R_B)
Control	ON/OFF or PWM

5.6.1 Connections and Control

Each LED channel is connected to a dedicated MCU GPIO pin through a series resistor that sets the maximum LED current:

$$I_{\text{LED}} \approx \frac{V_{\text{IO}} - V_f}{R} \tag{5.7}$$

where V_{IO} is the MCU I/O voltage, V_f is the LED forward voltage (dependent on color), and R is the chosen current-limiting resistor. PWM can be used to reduce brightness and average current while maintaining visibility.

5.6.2 Placement Considerations

The LED is placed to remain visible during use, while being kept away from the ambient light sensor field-of-view to avoid optical interference. LED traces are routed away from sensitive sensor lines to reduce coupling.

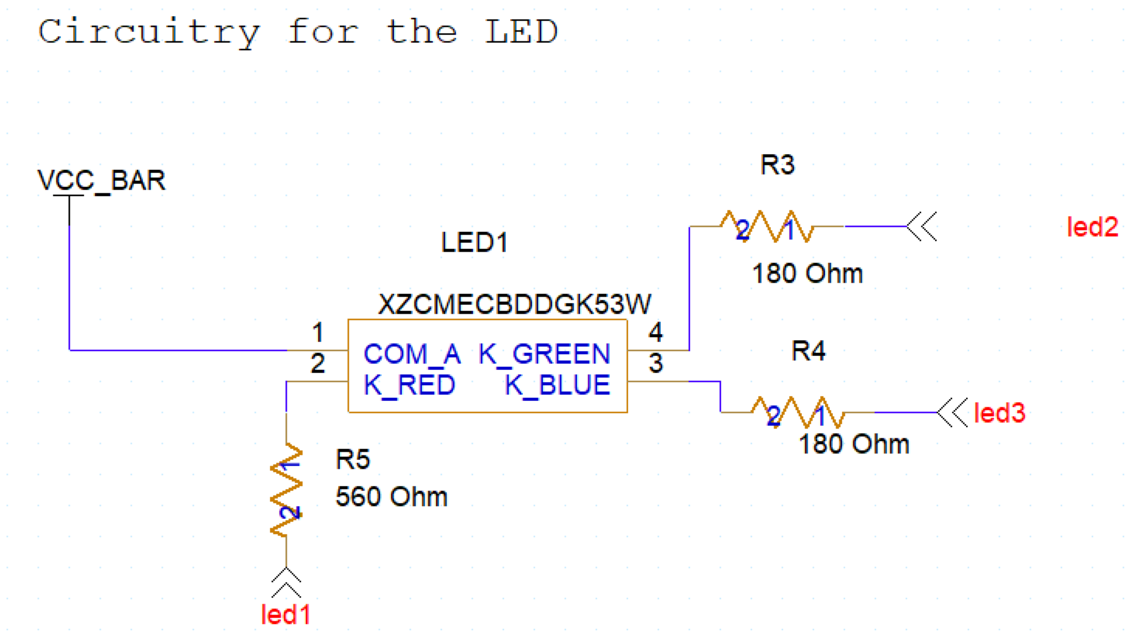

Figure 5.8: Status LED schematic.

5.6.3 Current-Limiting Resistor Calculation (Design Proof)

Each LED channel is driven through a series resistor to set the forward current. Assuming a regulated supply $V_{DD} = 3.00\,\text{V}$ and a target LED current $I_F = 2.00\,\text{mA}$, the resistor value is obtained from Ohm's law:

$$R = \frac{V_{DD} - V_F}{I_F} \tag{5.8}$$

where V_F is the forward voltage of the corresponding LED die.

Red channel (R5). Using an estimated red forward voltage $V_{F,\text{red}} \approx 1.80\,\text{V}$:

$$R_{\text{red}} = \frac{3.00\,\text{V} - 1.80\,\text{V}}{2.00\,\text{mA}} = \frac{1.20\,\text{V}}{2.00\,\text{mA}} = 600.00\,\Omega \tag{5.9}$$

The closest standard value was selected:

$$R5 = 560.00\,\Omega \tag{5.10}$$

which results in an actual current of:

$$I_{\text{red}} = \frac{3.00\,\text{V} - 1.80\,\text{V}}{560.00\,\Omega} \approx 2.14\,\text{mA} \tag{5.11}$$

Green/Blue channels (R3 and R4). Using an estimated forward voltage $V_{F,\text{green/blue}} \approx 2.65\,\text{V}$:

$$R_{\text{gb}} = \frac{3.00\,\text{V} - 2.65\,\text{V}}{2.00\,\text{mA}} = \frac{0.35\,\text{V}}{2.00\,\text{mA}} = 175.00\,\Omega \tag{5.12}$$

The closest standard value was selected:

$$R3 = R4 = 180.00\,\Omega \tag{5.13}$$

which results in an actual current of:

$$I_{\text{gb}} = \frac{3.00\,\text{V} - 2.65\,\text{V}}{180.00\,\Omega} \approx 1.94\,\text{mA} \tag{5.14}$$

5.7 Reset Button schematic

The button to be chosen for the asynchronous reset of the MCU is the GPTS203211B. which is a push-button. The main characteristics are summarized in the table 5.7 The circuit connections are represented in the figure 5.9

Table 5.7: Status LED main characteristics

Parameter	Value
RedDes	U5
Type	Push button
Part number	GPTS203211B
Contact resistance	100 mΩ
Insulation resistance	100 MΩ
Life cycle	20,000 cycles

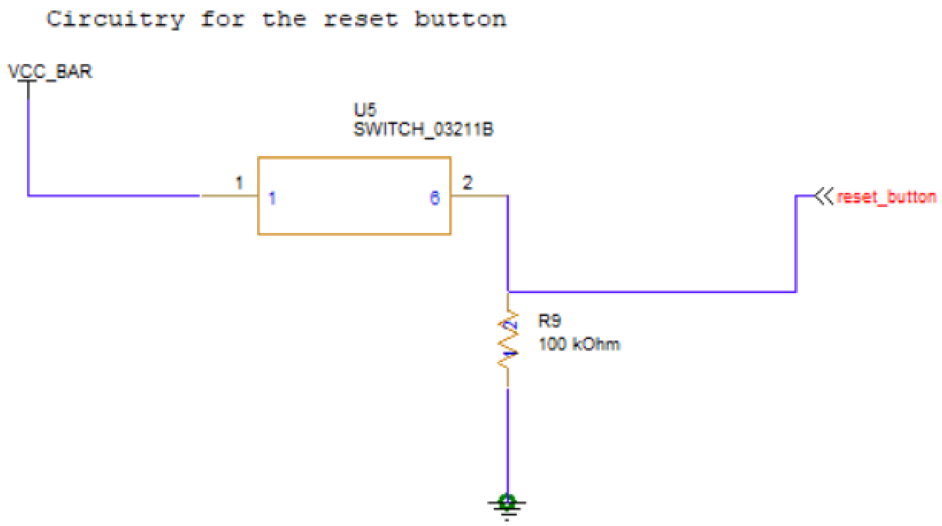

Figure 5.9: Reset Button Schematic

5.7.1 Reset Button Pull-Down Resistor (R9) Selection

R9 is used as a pull-down resistor on the `reset_button` signal to ensure a defined logic level when the switch is open, preventing the MCU reset input from floating and being triggered by noise. The value is chosen as a compromise between input stability and power consumption when the button is pressed.

When the button is pressed, current flows from `VCC_BAR` through the switch and R9 to ground:

$$I_{\text{press}} = \frac{V_{CC}}{R9} \tag{5.15}$$

Using $V_{CC} = 3.00\,\text{V}$ and $R9 = 100.00\,\text{k}\Omega$:

$$I_{\text{press}} = \frac{3.00\,\text{V}}{100.00\,\text{k}\Omega} = 30.00\,\mu\text{A} \tag{5.16}$$

which corresponds to a negligible power dissipation:

$$P_{\text{press}} = V_{CC} \cdot I_{\text{press}} \approx 90.00\,\mu\text{W} \tag{5.17}$$

Therefore, $100.00\,\text{k}\Omega$ provides a stable pull-down while keeping the button-press current very small.

Chapter 6

Bill of Materials and Cost Estimate

6.1 Bill of Materials (BOM)

Table 6.1: Bill of Materials (with unit and extended price). Prices are based on the project slides.

Item	**Qty**	**Reference**	**Part**	**Unit ()**	**Ext. ()**	**Notes**
1	1	ANT1	RFANT3216120A5T	0.30	0.30	Chip antenna
2	1	B1	USE-702528-500PCBJST (500 mAh)	5.49	5.49	Li-Po battery
21	1	U1	OPT3001DNP	1.98	1.98	Optical sensor
18	1	MTC1	ICS-43434	2.82	2.82	Acoustic sensor (I^2S mic)
25	1	U5	CC2640R2FRHB	4.71	4.71	BLE MCU
13	1	LED2	XZCMECBDDGK53W	0.62	0.62	RGB status LED
22	1	U2	TLV75801PDBVR	0.23	0.23	Adjustable LDO
24	1	U4	MAX1555EZKT	2.50	2.50	Battery charger
–	–	(Bundle)	Passive components + switch	3.51	3.51	C*, L*, FL1, R*, Y*, J*, SW*, etc.
–	1	(PCB)	4-layer PCB fabrication	5.19	5.19	Manufacturing
				TOTAL	**27.35**	Project total

6.2 Notes on Costing

Individual prices are taken from the project slides where explicitly provided. All remaining unpriced items in the BOM (capacitors, inductors, ferrite bead, crystals, connectors, resistor network, and switches) are grouped under the “Passive components + switch” bundle to match the reported total cost.

Chapter 7

Printed Circuit Board

7.1 General Specifications

Parameter	Value/Description
Number of Layers	2-layer
Board Dimensions	61.14 mm × 96.67 mm
Overall Thickness	[e.g., 1.55 mm]
Solder Mask Color	Green
Silkscreen Color	White
Eurocircuit pattern class	6
Eurocircuit drill class	b
Minimum Trace Width	0.15 mm
Trace Width for VDDS	0.2 mm
Minimum Spacing	0.15 mm
Minimum Via Diameter	0.25 mm
Via Type	PTH

Table 7.1: PCB Construction Specifications

7.2 Board Outline and Mounting

- **Board outline:** The PCB uses a rectangular outline with rounded corners and is dimensioned to mechanically host a Li-Po battery area (B1) on the right side. The outline also includes an RF edge region for the BLE antenna placement near the top-left corner.

- **Mounting holes:** The board includes mounting holes on the right side (and one near the center), allowing the PCB to be fixed to an enclosure using screws/spacers. Hole placement was chosen to provide mechanical stability while keeping clearances from sensitive routing and components.

- **Battery integration:** A dedicated keepout/footprint region is reserved for the battery pack (B1). The battery is intended to be secured to the board (or enclosure) using adhesive tape/foam pad, while its electrical connection is made through the battery connector.

- **User access (switch and reset):** The On/Off switch (SW1) is located at the bottom-left edge to be accessible from the enclosure exterior. The reset button is placed on the left side and can be accessed through a small opening (or used mainly during debugging, depending on enclosure design).

- **Connectors:** The board provides a programming/debug connector (JTAG) and a battery connector, placed near the board edge to simplify wiring and assembly.

7.3 Layout Figures

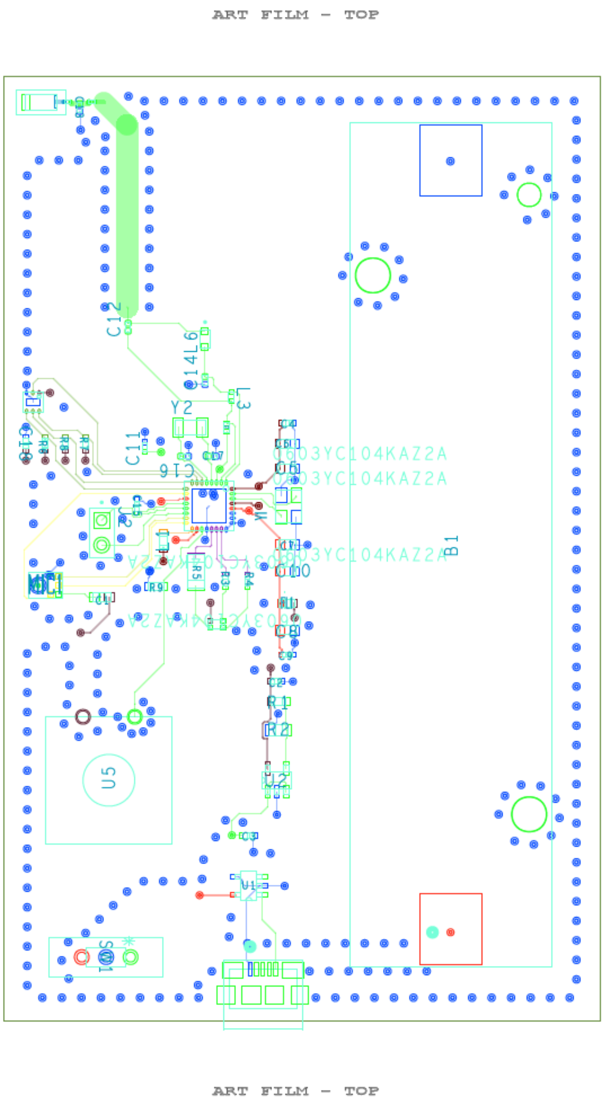

Figure 7.1: Layout Top

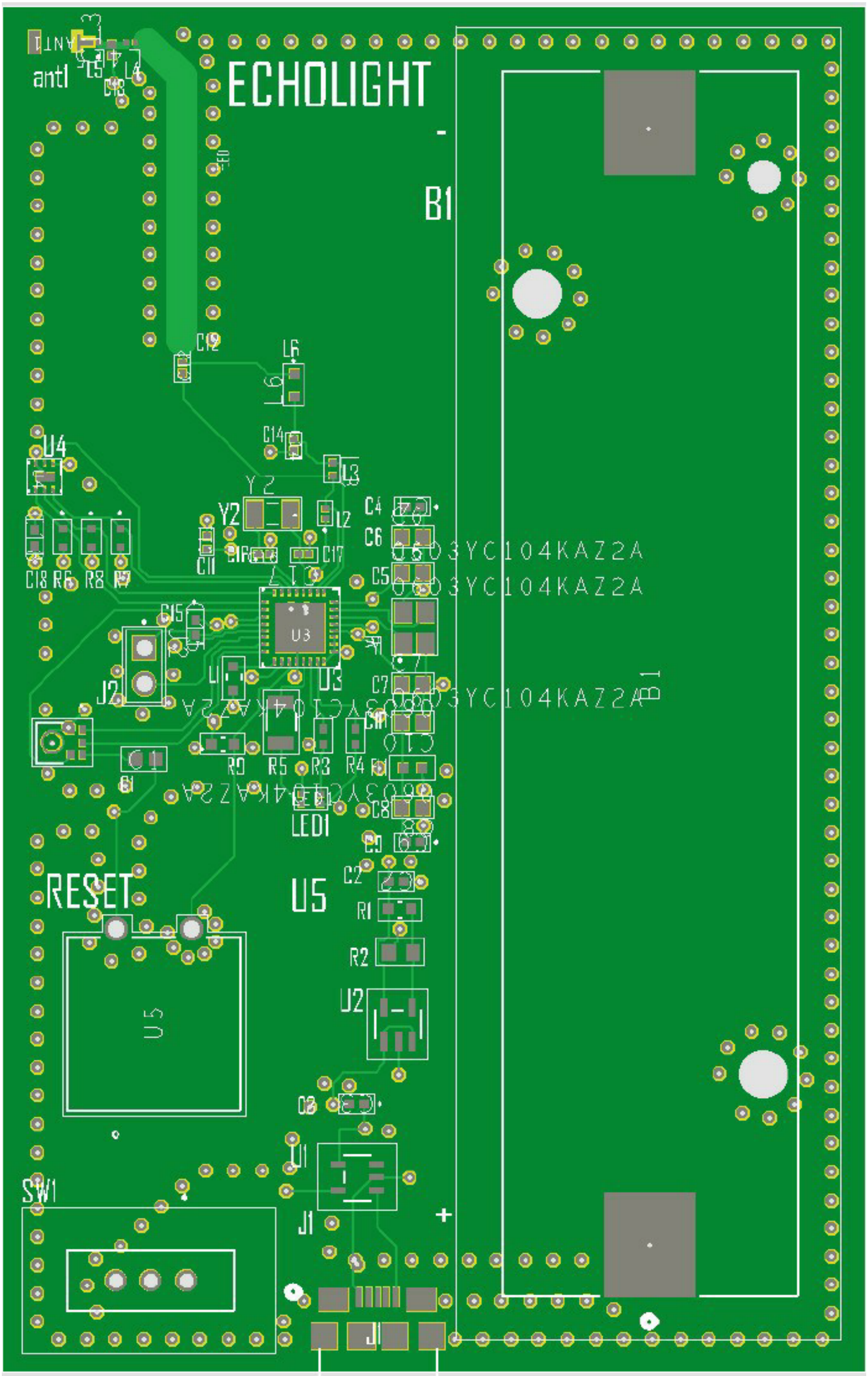

Figure 7.2: PCB Top

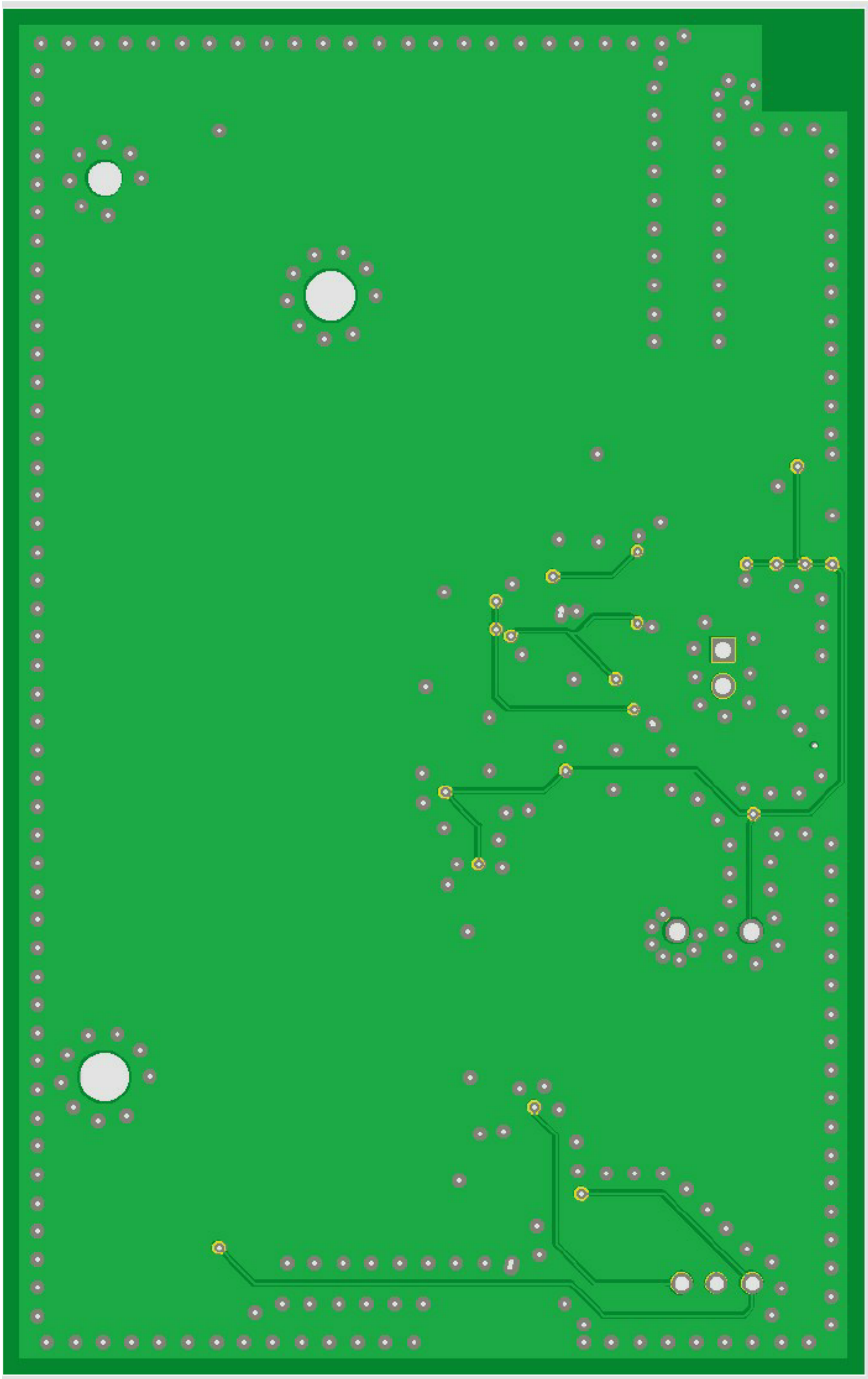

Figure 7.3: PCB Bottom

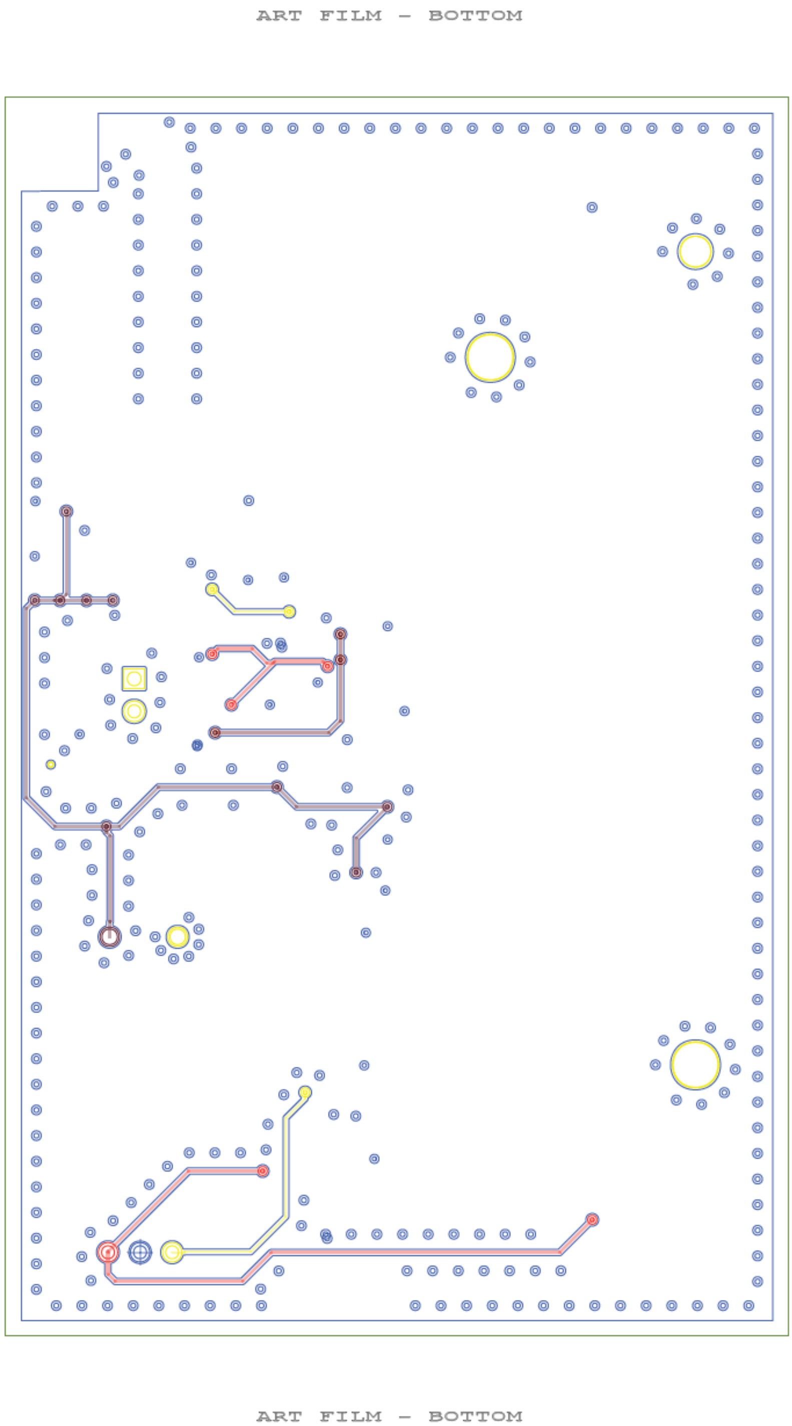

Figure 7.4: Layout Bottom

7.4 Placement Rationale

Component placement was driven by three priorities: (i) reliable BLE RF performance, (ii) minimal coupling between sensing channels and digital/RF activity, and (iii) short, stable power/ground return paths.

7.4.1 RF (MCU + Matching Network + Antenna)

The BLE MCU and the antenna network were placed as a compact RF block. The matching components were positioned immediately between the MCU RF pins and the antenna feed to minimize trace length and parasitic inductance/capacitance. The antenna was placed near the PCB edge and a keepout region was respected according to the antenna reference layout to reduce detuning and improve radiation efficiency. High-speed digital lines and switching power traces were kept away from the RF feed to limit interference.

7.4.2 Acoustic Channel (Digital MEMS Microphone)

The microphone placement and routing were chosen to minimize coupling from the RF section and from the power-management circuitry. The I^2S lines (clock/data) were routed as short as practical and kept separated from the RF feed and crystal traces. A local decoupling capacitor was placed close to the microphone supply pins to provide a low impedance return path at high frequency.

7.4.3 Optical Channel (Ambient Light Sensor)

The light sensor placement was chosen to reduce measurement bias by keeping it away from light-emitting components and from potential shadowing structures. I^2C pull-ups (if external) and the sensor decoupling capacitor were placed close to the sensor to maintain signal integrity and stable supply.

7.4.4 Power Management (Charger + LDO + Bulk Capacitors)

The battery charger and the LDO were placed near the power entry (USB connector and battery connector) to keep high-current paths short. Input and output capacitors were placed as close as possible to the charger/LDO pins to ensure stability and to reduce voltage ripple. The power switch was placed in series with the battery rail such that the OFF state fully disconnects the regulator input from the battery.

7.4.5 General Layout Rules

- Decoupling capacitors were placed close to each IC supply pin, with direct vias to the ground plane.
- Noisy blocks (RF, digital clocks, power switching paths) were separated from the sensing blocks (microphone and light sensor).
- Return paths were kept short and continuous by prioritizing solid ground planes and minimizing splits under sensitive traces.

7.4.6 Signal Integrity and Noise Mitigation (Key Layout Choices)

Signal integrity and interference reduction were treated as a primary design objective due to the need for stable, time-aligned multi-channel measurements and reliable BLE communication. The PCB layout therefore applies dedicated techniques to control return paths, reduce coupling, and limit switching noise.

- **Ground stitching vias:** stitching vias were placed along the RF feed line, around high-frequency signal routes, and near power traces to improve return-current continuity and reduce electromagnetic coupling. In addition, stitching vias were added around layer transitions (signal vias) to provide a nearby return path when signals change layers, limiting loop area and minimizing radiated and conducted noise.
- **Digital/analog separation:** sensing-related components and routes were kept physically separated from digital/RF activity where possible. Digital and analog/sensitive signal paths were routed to avoid crossing each other, and placement was chosen to prevent high-speed lines from passing through the sensing regions. This reduces crosstalk and prevents digital switching noise from degrading sensor measurements.
- **Simultaneous switching noise (SSN) mitigation:** To reduce simultaneous switching noise (SSN), I/O groups were supported with nearby power/ground connections and local decoupling, reducing ground bounce and supply droop during switching events.

Chapter 8

Project Plan (GANTT)